

Python in Excel: Your First Hour

Everything from today's workshop, on one page. Pin it next to your monitor.

1 Insert a PY() cell

- ▶ Ribbon: **Formulas** → **Insert Python**
- ▶ Type `=PY(` in any cell
- ▶ Shortcut: `Ctrl` `Alt` `Shift` `P`
- ▶ Run the cell with `Ctrl` `Enter`, not Enter alone
- ▶ Toggle output mode: `Ctrl` `Alt` `Shift` `M`

3 Profile a dataset in 30 seconds

- | | | |
|---------------------------------------|---|---|
| <code>.head()</code> first 5 rows | <code>.describe()</code> summary stats | <code>.columns</code> column names |
| <code>.tail()</code> last 5 rows | <code>.describe(exclude='number')</code> | <code>len(df)</code> row count |
| <code>.sample(5)</code> 5 random rows | <code>.isna().sum()</code> missing per column | <code>df['col']</code> single column (Series) |
| <code>.shape</code> (rows, columns) | <code>.dtypes</code> column types | <code>df[['a', 'b']]</code> several columns |

4 Count categories & group

```
sales_df['country'].value_counts()
sales_df['category'].value_counts(normalize=True)
```

```
# PivotTable, in code
sales_df.groupby('country')['unit_price'].mean()
sales_df.groupby(['country', 'category'])['unit_price'].mean()
sales_df.groupby('country')['unit_price'] \
    .agg(['mean', 'min', 'max'])
```

2 Load a table → DataFrame

- ▶ Convert your range to an Excel Table first (`Ctrl` `T`) and name it
- ▶ `xl()` is the bridge from worksheet to Python

```
sales_df = xl("sales[#All]", headers=True)
sales_df          # preview as a card
```

5 Filter, sort, chain

```
# Filter rows
us = df[df['country'] == 'US']
```

```
# Sort, highest first
df.sort_values('unit_price',
               ascending=False)
```

```
# Filter → group → sort
(df[df['country'] == 'US']
 .groupby('category')['unit_price']
 .mean()
 .sort_values(ascending=False))
```

6 One chart that earns gasps

```
sales_df['revenue'] = sales_df['unit_price'] * sales_df['quantity']

import seaborn as sns
sns.heatmap(sales_df.corr(numeric_only=True),
            annot=True, cmap='RdBu_r', center=0)

# Bonus, every numeric variable vs. every other
sns.pairplot(sales_df, hue='category')
```

⚠ THE CALC-ORDER RULE

Python in Excel runs cells **left-to-right, top-to-bottom**, like reading a page. If you define a variable in **B2**, you can reference it in any cell to the right in row 2, or in any row below. Referencing it from A1 returns **#PYTHON**. **Recovery rule:** put imports and variables in the top-left, build down and right.

THREE TO REMEMBER

1. Cells run top-left → bottom-right. 2. Cloud-only: no internet, no PY(). 3. Default output is Python Object, toggle to Excel Value to spill.

WHERE NEXT

Docs: learn.microsoft.com (search "Python in Excel").
Go deeper: *Advancing into Analytics* (O'Reilly).

DEFAULT ENVIRONMENT

pandas, numpy, scipy, matplotlib, seaborn, statsmodels, scikit-learn. No internet packages or local file I/O.